

ML4T Project 1: Martingale

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Introduction:

This Report describes my experiment with the American roulette wheel game using computational simulation. The simulation adapts the betting strategy of Professor Balch described in the project wiki:

<https://gatech.instructure.com/courses/430880/assignments/1952092>. Based on my real American roulette wheel game experience I will set the winning possibility of single spin as 18/38 in the following experiment. I will describe those experiment results in detail and provide the necessary mathematical calculations to prove my experiment result.

Question 1:

In Experiment 1, the estimated probability of winning \$80 within 1000 sequential bets is nearly 100%. In this experiment, we assume that the player has an unlimited bankroll, so we can calculate the probability using math.

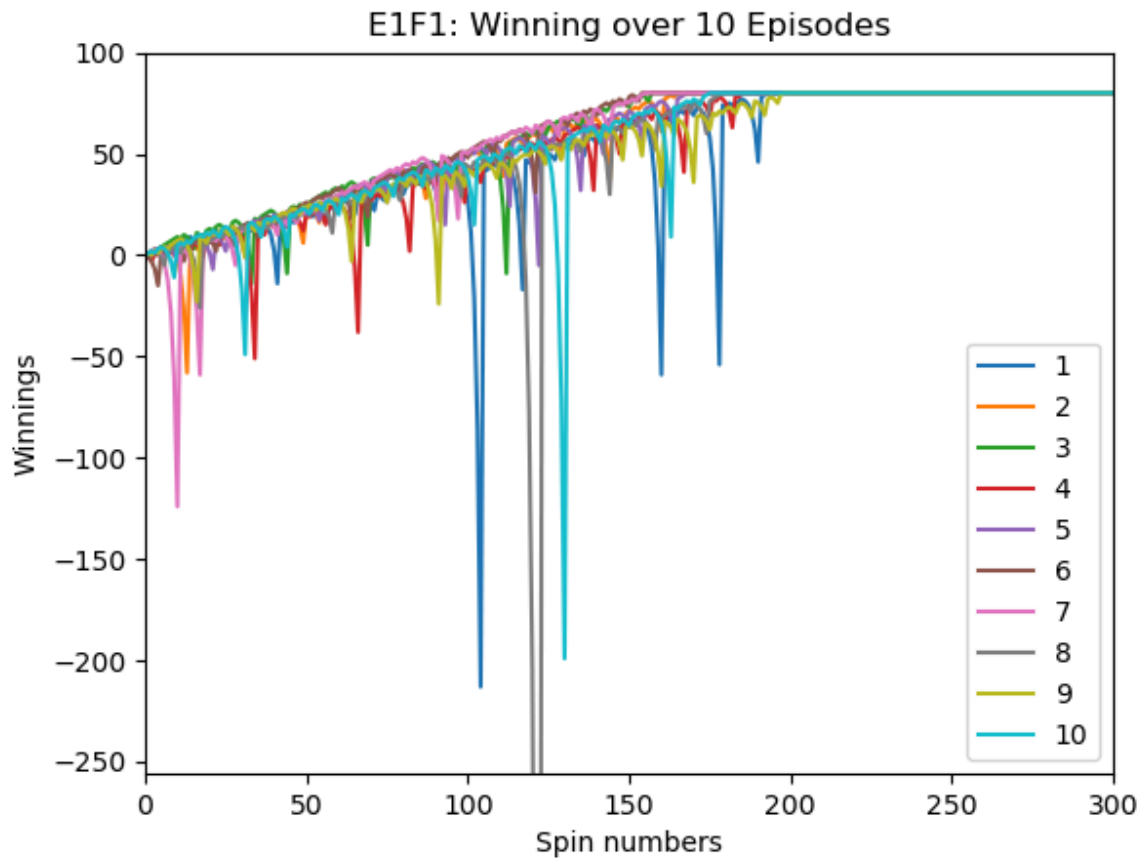
The probability of winning a single spin is 9/19. With each win, the player accumulates \$1 compared to its last win. So, to accumulate \$80, the player only needs to win more than 80 bets within the 1000 sequential bets.

Probability of winning less than 80 times within 1000 is :

$$P(<80) = \sum_{k=1 \text{ to } 79} nCr(1000, k) * (9/19)^k * (10/19)^{(1000-k)} \ll 0.0001$$

So the probability of winning \$80 is 1- P(<80) = 100%

The experiment figure E1F1 also proves the calculation: all 10 episodes finally reached \$80 after 200 spins

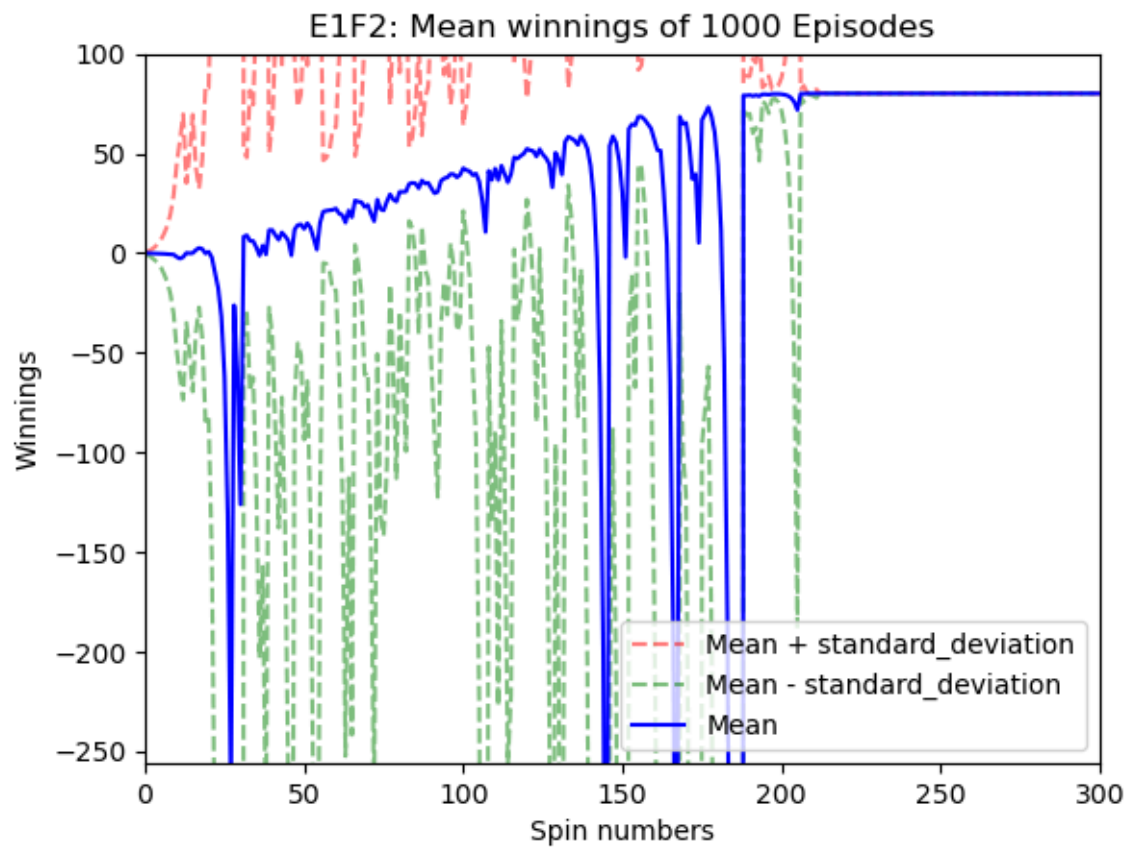


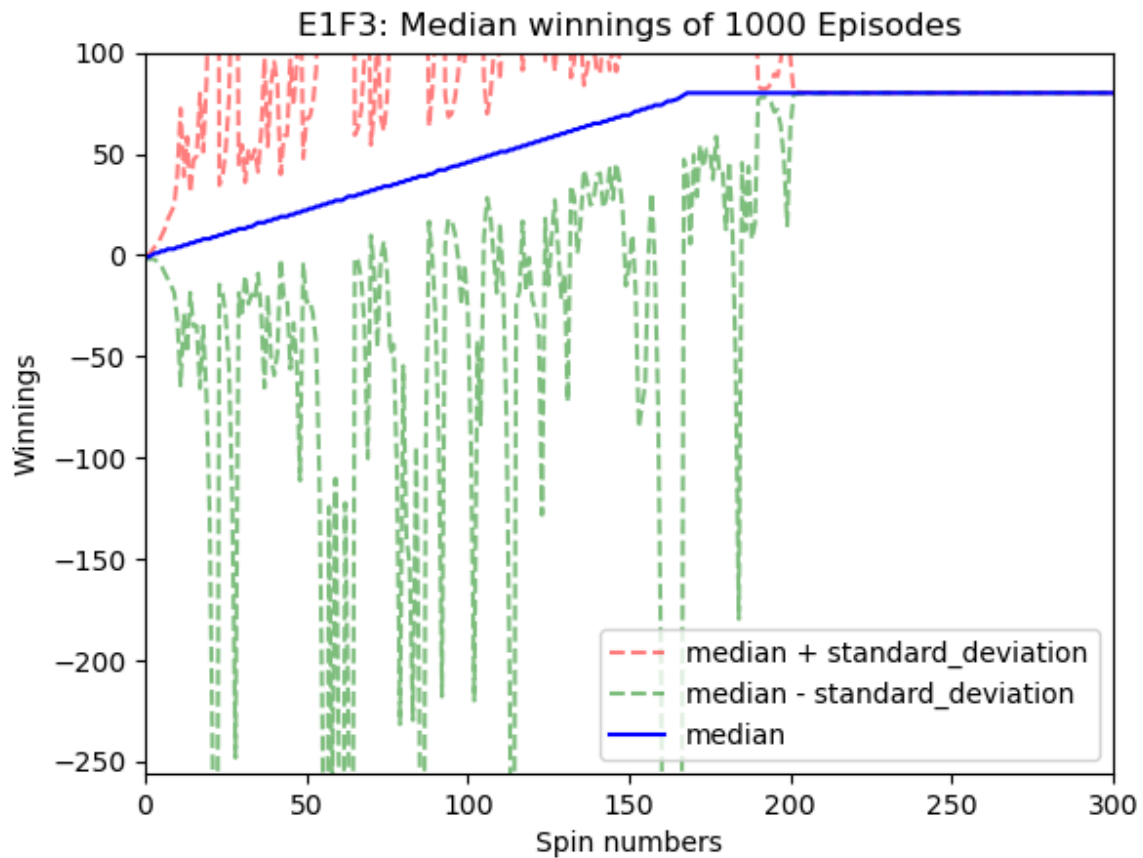
Question 2:

The estimated expected value of winning after 1000 sequential bets is \$80. As we calculated in Q1, the probability of winning less than \$80 is $P(<80) \ll 0.0001$. and the winnings stop accumulating after it reaches \$80, as we have defined in the algorithm. So the ultimate number of winning is \$80.

Question 3:

As the figures E1F2 and E1F3 show, the upper and lower standard deviation of mean and median will finally converge to \$80 after around 200 spins. Because all of the episodes will eventually reach \$80 and thus the standard deviation of the mean and median will become 0. The data becomes less and less variable, as a result, the three lines will finally overlap when all of the other episodes' winnings reach \$80.



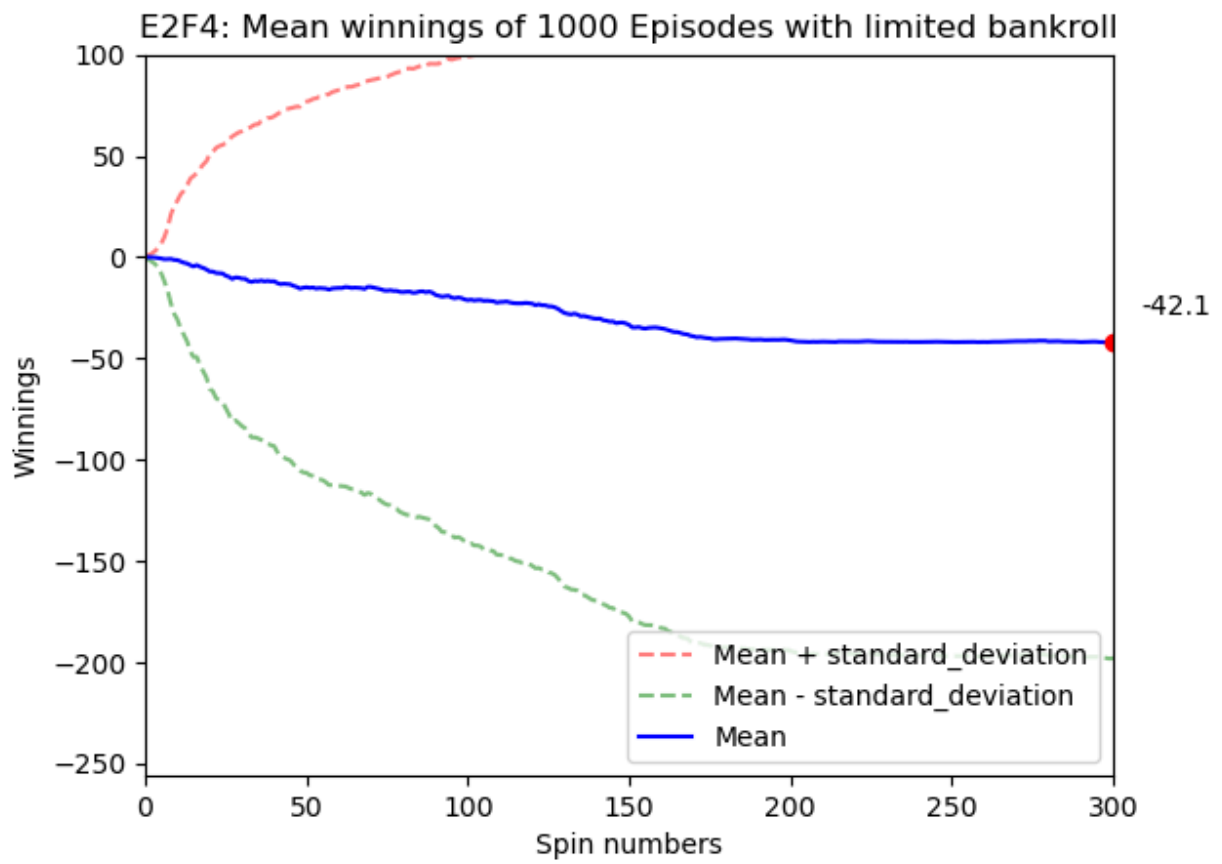


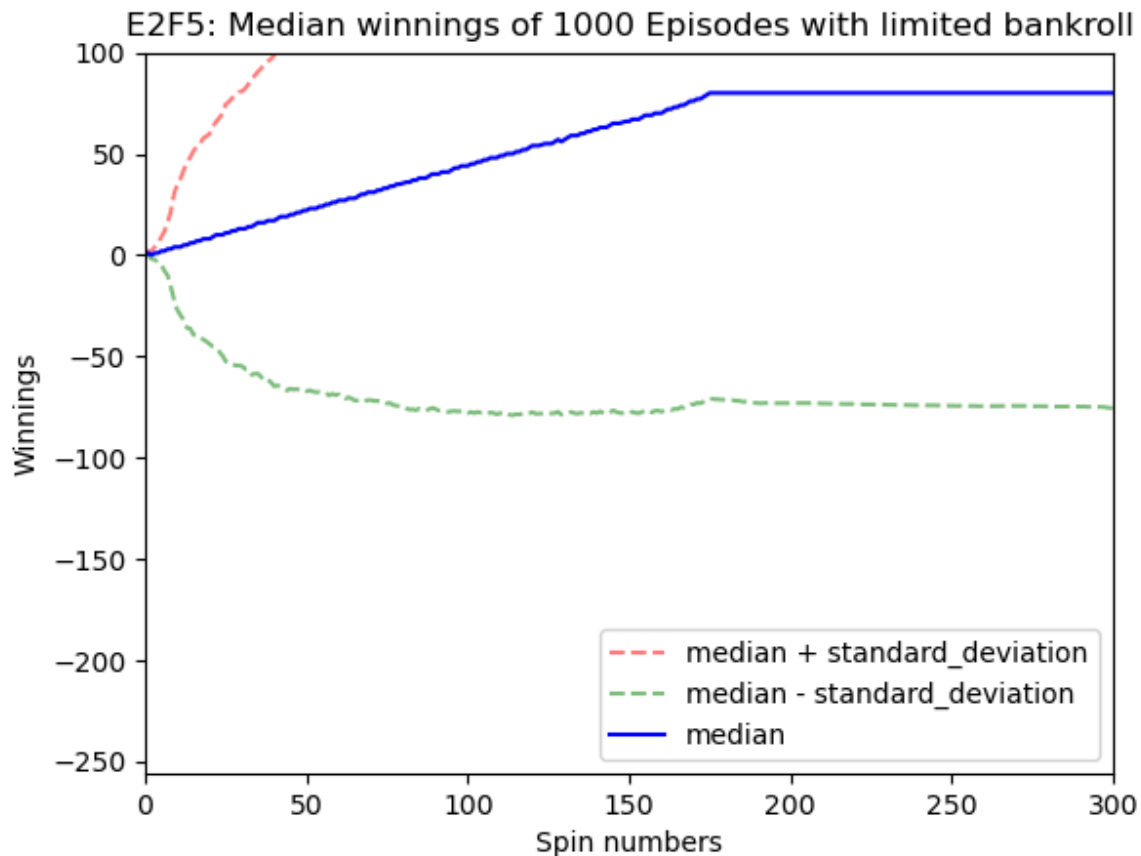
Question 4:

In experiment 2, the meaning value is stable to be around \$-42.1 after 200 spins. All three lines in both Figure 4 and Figure 5 are getting stabilized. Also according to Experiment 1, the result after 1000 bets only has two options: won (\$80) or loss (\$-256). We can calculate the win probability P based on below formula:

$$80 * P - 256 * (1-P) = -42.1$$

Then we get $P = 0.64$, so the probability of winning after 1000 sequential bets is 64%.





Question 5:

In Experiment 2, the estimated expected value of winnings after 1000 sequential tries is the same as the mean value, which is -42.1. As the previous questions proved, the result in 1000 tries is either \$80 or -\$256. So the expected value calculation is the same as the mean value.

Expected Value after 1000 tries: $64\% \cdot 80 - 36\% \cdot 256 = \-42.1 .

Figure E2F4 has the value marked as well.

Question 6:

Yes, the upper standard deviation line and lower standard deviation line will stabilize at a maximum (minimum) value. As the experiment 2 figures (E2F4, E2F5) show, the deviation

lines are eventually stabilized and are parallel to the mean | median line. After 200 tries, most of the results either reach \$80 or \$-256. and their proportion of each situation will not change after that. So the deviation becomes a fixed number. Since the mean | median number is stable, the value of deviation lines is also stable and stays in the minimum | maximum value.

Question 7:

Expected values can mathematically represent the average result of the experiment, while one result from a specific random seed can only represent this specific result, and this might be different for the next attempts with a different seed. The random sample might be misleading and cause potential bias if we use this result for other calculations. The expected value is more convincing to describe a problem with probability and avoid uncertainty.